What Could Be Built: Seven Engineering Frontiers Suggested by the Mathematics of the Cathedral

Jason Robert Gochanour

April 19, 2026

Abstract

The Cathedral establishes a machine-verified equivalence between the Riemann Hypothesis and the decay of the Nyman–Beurling distance, using techniques from spectral theory, harmonic analysis, variational methods, and number-theoretic sieve constructions. While the proof itself concerns pure mathematics, the *structures* it reveals—arithmetic Gram matrices, Möbius sieve weights, spectral gaps over multiplicative lattices, and the Parseval Bridge between time and frequency domains—suggest concrete engineering possibilities that do not currently exist. This paper proposes seven grounded, physically realizable technologies that could be built by adapting the Cathedral's mathematical insights to engineering domains: (1) prime-spaced sensor arrays with number-theoretic sidelobe control, (2) arithmetic compressed sensing with provable recovery guarantees, (3) Möbius-weighted adaptive filters for harmonic signals, (4) machine-verified stability certificates for safety-critical control systems, (5) prime-gap metamaterials with quasi-random electromagnetic properties, (6) arithmetic quantum error correction codes, and (7) formally verified numerical solvers with axiom-tiered trust. For each proposal, we describe the existing technology, the mathematical connection to the Cathedral, the engineering design, and the feasibility assessment.

Contents

1 Introduction: From Proof to Possibility

The companion paper *From Zeta Zeros to Engineering* mapped the Cathedral's mathematics to existing engineering concepts—a dictionary. This paper asks a different question: *what new things could be built?*

The proposals here are speculative but grounded. Each satisfies three criteria:

1. **Mathematical foundation**: The proposal is directly connected to a mathematical structure in the Cathedral (not merely inspired by it).

2. **Physical realizability**: The proposed device or algorithm could be built with current or near-term technology.
3. **Novel capability**: The proposal offers a capability that does not exist in current engineering practice.

We are explicit about what is established, what is plausible, and what is speculative. The Cathedral proves theorems; this paper only suggests possibilities.

2 Prime-Spaced Sensor Arrays

2.1 What Exists Today

Phased array antennas and sonar arrays use either:

- **Uniform spacing** ($\lambda/2$): Simple, but produces periodic grating lobes that limit the usable bandwidth.
- **Random spacing**: Eliminates grating lobes but produces unpredictable sidelobe levels, requiring more elements for reliable performance.

Neither fully solves the fundamental tradeoff: predictable pattern control *and* wide bandwidth.

2.2 What the Cathedral Suggests

The Gram matrix G_N encodes the correlation structure of the basis functions $\{1/(kx)\}$ for $k = 2, \ldots, N$. Its entries depend on $\gcd(j, k)$ and $\mathrm{lcm}(j, k)$—purely arithmetic quantities. The key insight: this arithmetic structure is neither periodic (like uniform spacing) nor random, but *quasi-random with multiplicative structure.*

The Selberg sieve weights $w_k = \mu(k)(1 - \ln k/ \ln N)^+$ provide the optimal beamforming coefficients for this arithmetic array.

2.3 The Engineering Design

Proposal: A sensor array with element positions at $x_k = k \cdot d$ for $k = 2, 3, 5, 7, 11, \ldots, p_N$ (prime positions only), with beamforming weights derived from the Selberg sieve.

Properties (conjectured from the mathematics):

- No grating lobes (primes are not periodic).
- Predictable sidelobe behavior (controlled by the Gram matrix condition number $\kappa \sim N \ln N$).
- Optimal weights computable via the Rayleigh–Ritz principle applied to the arithmetic Gram matrix.
- The Parseval Bridge provides an exact relationship between the spatial pattern and the frequency response.

2.4 Feasibility

High. Non-uniform arrays are already used in radio astronomy (ALMA, SKA) and military radar. The mathematical computation (Gram matrix, Selberg weights) is straightforward for $N \leq 10{,}000$ elements. The novel contribution is using number-theoretic structure (primes, Möbius function) instead of optimization-based aperiodic thinning.

Required development: Simulation of prime-spaced array patterns; comparison with conventional thinning algorithms; experimental validation with a small ($N \sim 100$) prototype.

3 Arithmetic Compressed Sensing

3.1 What Exists Today

Compressed sensing recovers sparse signals from fewer measurements than the Nyquist rate requires, using random measurement matrices that satisfy the Restricted Isometry Property (RIP). The standard approach: use i.i.d. Gaussian or Bernoulli matrices, which satisfy RIP with high probability.

The limitation: random matrices are expensive to store and slow to apply. Structured matrices (Fourier, Hadamard) are faster but have weaker RIP guarantees.

3.2 What the Cathedral Suggests

The fractional-part functions $\{1/(kx)\}$ form a structured, deterministic basis on $(0, 1)$ with coherence controlled by $\gcd(j, k)$. The Gram matrix G_N is the "sensing coherence matrix," and the Cathedral proves its minimum eigenvalue is strictly positive for all N—exactly the condition needed for stable recovery.

3.3 The Engineering Design

Proposal: An arithmetic sensing matrix $A_{mk} = \{1/(kx_m)\}$ where x_m are measurement points drawn from $(0, 1)$ and $k = 2, \ldots, N$ indexes the "arithmetic dictionary."

Recovery guarantee: If the signal is s-sparse in the arithmetic dictionary, then $O(s \cdot \ln N)$ measurements suffice for recovery via ℓ_1 minimization, because the Gram matrix eigenvalue bound $\lambda_{\min}(G_N) \geq C/\ln N$ provides a coherence bound.

Advantages:

- Deterministic (no randomness needed).
- Fast matrix-vector products via FFT-like algorithms exploiting the multiplicative structure of the entries.
- The sensing matrix has arithmetic structure that natural signals (audio, vibration) may already possess: harmonic overtones at integer multiples of a fundamental frequency are exactly the signals that are sparse in this dictionary.

3.4 Feasibility

Medium. The mathematical ingredients (Gram matrix eigenvalue bounds, coherence estimates) exist in the Cathedral. What's needed: (1) a formal RIP proof for the arithmetic sensing matrix, (2) comparison with Gaussian and Fourier sensing on synthetic and real signals, (3) identification of signal classes that are naturally sparse in the arithmetic dictionary.

Most promising application: Vibration monitoring in rotating machinery, where the signal is naturally a sum of harmonics at integer multiples of the rotation frequency.

4 Möbius-Weighted Adaptive Filters

4.1 What Exists Today

Adaptive filters (LMS, RLS, Kalman) adjust coefficients to minimize mean squared error, typically using uniformly weighted or exponentially weighted observations. These work well for signals with additive structure (sum of sinusoids, noise).

Signals with *multiplicative* structure—harmonic overtones, frequency combs, musical signals—are poorly matched by standard additive models. The harmonic sieve (identifying which overtones are present) is a combinatorial problem.

4.2 What the Cathedral Suggests

The Cathedral's optimal approximation uses Möbius-weighted coefficients: $v_k = \mu(k) \cdot w(k)$, where $\mu(k)$ is the Möbius function and $w(k)$ is a tapering window. The Möbius function naturally encodes multiplicative structure:

- $\mu(k) \neq 0$ only for squarefree k (fundamental harmonics, no repeated prime factors).
- The Möbius inversion formula $\sum_{d|n} \mu(d) = [n = 1]$ provides exact extraction of fundamental components from a harmonic mixture.

4.3 The Engineering Design

Proposal: An adaptive filter whose weight update rule incorporates Möbius structure:

$$w_k^{(n+1)} = w_k^{(n)} + \alpha \cdot \mu(k) \cdot e(n) \cdot x_k(n),$$

where $\mu(k)$ pre-weights the update by the Möbius function, naturally suppressing composite harmonics and enhancing fundamental-frequency estimation.

Applications:

- **Musical pitch detection**: Identify the fundamental frequency when only overtones are present (the "missing fundamental" problem).
- **Vibration diagnostics**: Separate bearing defect frequencies from shaft harmonics in rotating machinery.

- **Power grid monitoring**: Identify harmonic distortion sources in electrical power systems, where harmonics at multiples of 50/60 Hz corrupt the signal.

4.4 Feasibility

High. The Möbius function is trivially precomputable for $k \leq 10^6$ (sufficient for audio and vibration). The filter structure is a minor modification to existing LMS algorithms. Prototype implementation requires only a few hundred lines of code.

Required validation: Comparison with standard LMS on synthetic harmonic signals; testing on real audio and vibration datasets; analysis of convergence properties of the Möbius LMS.

5 Machine-Verified Stability Certificates

5.1 What Exists Today

Safety-critical systems (aircraft flight control, nuclear reactor control, autonomous vehicles) require stability proofs. These are typically:

- Pen-and-paper Lyapunov arguments, checked by human review.
- Simulation-based validation (not a proof).
- Sum-of-squares (SOS) programming for polynomial systems, verified by numerical solvers with floating-point errors.

None of these provide the level of assurance that a machine-checked proof offers.

5.2 What the Cathedral Suggests

The Cathedral proves positive definiteness of the Gram matrix H_N for all $N \geq 1$—a family of stability certificates, parameterized by N, verified by the Lean 4 type-checker. The proof architecture (axiom registry, tiered assumptions, `#print axioms` discipline) provides a template for formally verified stability proofs.

5.3 The Engineering Design

Proposal: A framework for machine-verified Lyapunov stability certificates, structured like the Cathedral:

1. **Define** the system dynamics $\dot{x} = f(x)$ in Lean 4.
2. **Define** a candidate Lyapunov function $V(x)$.
3. **Prove** $V(x) > 0$ for $x \neq 0$ (positive definiteness).
4. **Prove** $\dot{V}(x) < 0$ along trajectories (decrease condition).

5. **Axiomatize** any numerical estimates (e.g., bounds on nonlinear terms) as named, tracked axioms.

6. **Audit** the axiom registry to identify which stability guarantees depend on which numerical assumptions.

The "axiom tiering" system from the Cathedral directly applies: Tier 1 axioms (kernel) are trusted; Tier 2 axioms (numerical bounds) are verified by interval arithmetic; Tier 3 axioms (model assumptions) are validated by testing.

5.4 Feasibility

Medium-high. Lean 4 already has the mathematical infrastructure (Mathlib's analysis library). The gap is domain-specific: we need formalized control theory definitions (transfer functions, state-space models, Lyapunov functions) and connections to numerical verification tools. The Cathedral provides the organizational template.

Impact: A formally verified autopilot stability proof would be a major advance in aviation safety certification. Current DO-178C standards require extensive testing; a machine-checked proof would complement or replace testing for the most critical properties.

6 Prime-Gap Metamaterials

6.1 What Exists Today

Metamaterials are engineered structures with electromagnetic properties not found in nature: negative refractive index, perfect absorption, cloaking. Their properties depend on the geometry of sub-wavelength resonant elements arranged in patterns (usually periodic or graded).

Periodic arrangements produce band gaps (photonic crystals). Disordered arrangements produce broadband absorption. The tradeoff between narrow-band performance (periodic) and broadband performance (disordered) is a fundamental design constraint.

6.2 What the Cathedral Suggests

The prime numbers are neither periodic nor random—they are quasi-random with controlled statistics. The Montgomery–Odlyzko law shows that zeta zero spacings follow GUE statistics, and the Cathedral's Gram matrix encodes the correlation structure of prime-indexed basis functions.

Key insight: A metamaterial with resonant elements at prime-indexed positions would have:

- No Bragg peaks (primes are not periodic)→ no narrow-band resonance artifacts.
- Controlled density ($\sim 1/\ln N$ primes near N)→ predictable broadband absorption.
- GUE-like level repulsion in the resonant spectrum→ minimum spacing between resonant frequencies, preventing destructive interference.

6.3 The Engineering Design

Proposal: A broadband electromagnetic absorber consisting of resonant elements (split-ring resonators, patch antennas) with:

- Element sizes a_k proportional to k for $k = 2, 3, 5, 7, 11, \ldots$ (prime indices only).
- Each element resonates at frequency $f_k \propto 1/a_k$, producing a "harmonic comb" at prime frequencies.
- The overall absorption spectrum $A(f)$ is a superposition of resonances at prime-indexed frequencies, with the Nyman–Beurling theory providing theoretical bounds on the spectral coverage.

The conjecture: RH implies that this prime-frequency absorber achieves $A(f) \to 1$ (perfect absorption) as the number of elements $\to \infty$, because $d_N^2 \to 0$ measures exactly how well the prime-indexed resonances cover the unit interval.

6.4 Feasibility

Medium. Fabrication of metamaterials with non-uniform element sizes is routine (laser cutting, 3D printing). The novel contribution is the prime-indexed design rule and the theoretical connection to the Nyman–Beurling criterion. Simulation (CST Microwave Studio, COMSOL) should precede fabrication.

Key validation: Simulate the absorption spectrum of a prime-spaced resonator array and compare with (a) uniform spacing, (b) random spacing, (c) optimized aperiodic spacing. If the prime-spaced design offers competitive broadband performance with a closed-form design rule (no optimization needed), it would be practically useful.

7 Arithmetic Quantum Error Correction

7.1 What Exists Today

Quantum error correction (QEC) protects quantum information using stabilizer codes (surface codes, color codes, toric codes). These are built from algebraic structures (groups, lattices) over finite fields $\mathbb{F}_2$.

Current QEC codes do not exploit number-theoretic structure. Their design is based on topology (surface codes) or algebraic geometry (Goppa codes), not on multiplicative number theory.

7.2 What the Cathedral Suggests

The Gram matrix G_N defines an inner product on the space of arithmetic functions. Its positive definiteness (proved in the Cathedral) means that the arithmetic functions $\{1/(kx)\}$ are linearly independent in $L^2(0, 1)$—they form a valid code basis.

The Möbius function provides a natural "error syndrome": $\mu(n) = 0$ when n has a squared prime factor (an "error"), and $\mu(n) = \pm 1$ when n is squarefree ("no error"). The Möbius inversion formula provides exact error decoding.

7.3 The Engineering Design

Proposal: A family of quantum codes indexed by N, where:

- Code qubits correspond to prime indices $p \leq N$.
- Stabilizer generators correspond to composite numbers $n = p \cdot q$ (products of two primes).
- Error syndromes are decoded via Möbius inversion: the error pattern on composites determines the error on primes.
- The code distance is bounded by the minimum eigenvalue $\lambda_{\min}(G_N) \geq C/\ln N$, giving distance $\Omega(\ln N/\ln\ln N)$.

7.4 Feasibility

Low-medium. This is the most speculative proposal. The mathematical connection (Gram matrix $\to$ code distance) is rigorously grounded, but translating from real-valued L^2 inner products to binary quantum stabilizers requires a discretization step that may destroy the number-theoretic structure.

Required research: Formalize the connection between $L^2(0,1)$ positive definiteness and stabilizer code distance; construct explicit small codes ($N = 30, 50$) and compute their parameters; compare with surface codes of similar size.

8 Formally Verified Numerical Solvers

8.1 What Exists Today

Numerical PDE solvers (finite element, finite difference, spectral methods) are validated by:

- Convergence tests on known analytical solutions.
- Mesh refinement studies.
- Code-to-code comparisons.

None of these constitute a *proof* that the solver produces correct results for a new problem. The gap between "tested" and "verified" is the gap between engineering and mathematics.

8.2 What the Cathedral Suggests

The Cathedral's architecture—axioms at the boundary, machine-verified inference in the interior—is directly applicable to numerical methods:

1. **Axioms as model assumptions**: "The material is linear elastic" or "The flow is incompressible" are stated as formal axioms.

2. **Discretization as theorem**: "The FEM solution converges to the PDE solution as $h \to 0$" is proved from the axioms using Céa's lemma and interpolation estimates.

3. **Numerical bounds as tiered axioms**: "The floating-point implementation satisfies $|x_h - x_{\text{exact}}| < \varepsilon$" is a Tier 2 axiom, validated by interval arithmetic.

4. **Audit trail**: `#print axioms solution_valid` shows exactly which assumptions the final result depends on.

8.3 The Engineering Design

Proposal: A numerical solver framework in Lean 4 where:

- The PDE, boundary conditions, and material model are formalized as types.
- The discretization (mesh, shape functions, quadrature) is formalized with proved convergence.
- The linear algebra (assembly, solve) uses verified implementations or axiomatized routines with error bounds.
- The output is a solution *and* a machine-checked certificate that the solution satisfies the stated error bound, conditional on the named axioms.

8.4 Feasibility

Medium. The Cathedral demonstrates that 20,000 lines of nontrivial mathematics can be machine-verified in Lean 4. A simple 1D FEM solver with formal convergence proof would be ~5,000 lines—well within current capabilities. The challenge is scaling to 2D/3D problems and connecting to efficient compiled linear algebra.

Near-term milestone: A formally verified 1D Poisson solver with proved $O(h^2)$ convergence, producing a PDF certificate that could be submitted as part of an engineering design review.

9 Feasibility Summary

#	Proposal	Feasibility	Timeline
1	Prime-spaced sensor arrays	High	1–2 years
2	Arithmetic compressed sensing	Medium	2–4 years
3	Möbius adaptive filters	High	<1 year
4	Verified stability certificates	Medium-high	2–3 years
5	Prime-gap metamaterials	Medium	3–5 years
6	Arithmetic quantum error correction	Low-medium	5–10 years
7	Verified numerical solvers	Medium	2–4 years

The most immediately actionable proposals are **3** (Möbius adaptive filters—a software change to existing LMS algorithms) and **1** (prime-spaced arrays—a design rule for existing hardware). Both could be prototyped in months.

The highest impact proposals are **4** (verified stability certificates—transformative for safety-critical systems) and **7** (verified solvers—a new standard for engineering analysis trust). Both require significant Lean 4 library development but build directly on the Cathedral's organizational template.

10 What This Paper Is Not Claiming

For intellectual honesty:

1. We do not claim that the Riemann Hypothesis must be proved for any of these technologies to work. The mathematical structures (Gram matrices, Möbius weights, Parseval's theorem) exist independently of RH.

2. We do not claim that these technologies require the Cathedral. An engineer could discover any of these ideas independently. The Cathedral's contribution is *conceptual*: it demonstrates these structures working together in a verified framework.

3. We do not claim immediate commercial viability. Each proposal requires significant research, development, and validation before deployment.

4. We do not claim novelty in the component techniques (non-uniform arrays, compressed sensing, adaptive filters). The novelty is in the *arithmetic* structure imposed by number theory, and in the *verification* framework inspired by the Cathedral's proof architecture.

11 Conclusion: The Primes as Engineering Material

The primes are not a material that engineers can hold. They are not a force that can be measured. They are not a signal that can be recorded. But they are a *structure*—a pattern so deep that it appears in nuclear energy levels, random matrix theory, quantum chaos, and now in machine-verified proof architectures.

This paper suggests that the prime number structure can be treated as *engineering material*: a design principle for sensor arrays, a compression dictionary for harmonic signals, a window function for spectral analysis, and a template for verified computation.

The Cathedral did not prove the Riemann Hypothesis. But in the process of formalizing the attempt, it assembled a toolkit of mathematical structures that engineers have never used—not because the structures are unknown, but because the connection between number theory and engineering has never been made explicit.

These seven proposals are that connection. Some will fail. Some may succeed. All are grounded in real mathematics, verified by machine.

The primes are not a bridge you can walk across.
They are the blueprint for bridges not yet imagined.

References

[1] E. Candès, J. Romberg, and T. Tao, *Robust uncertainty principles: exact signal recovery from highly incomplete frequency information*, IEEE Trans. Inform. Theory **52** (2006), 489–509.

[2] S. Haykin, *Adaptive Filter Theory*, 5th ed., Pearson, 2014.

[3] H. Khalil, *Nonlinear Systems*, 3rd ed., Pearson, 2015.

[4] N. Engheta and R.W. Ziolkowski, *Metamaterials: Physics and Engineering Explorations*, Wiley, 2006.

[5] D. Gottesman, *Stabilizer Codes and Quantum Error Correction*, PhD thesis, Caltech, 1997.

[6] J.S. Hesthaven, *Numerical Methods for Conservation Laws*, SIAM, 2018.

[7] J.R. Gochanour, *From Zeta Zeros to Engineering: What the Cathedral's Mathematical Physics Means for Practicing Engineers*, 2026.